

Report on CVT (Continuously Variable Transmission) Model

1. Project Overview

Our project focuses on constructing a working model of a Continuously Variable Transmission (CVT) system. The CVT is a type of automatic transmission that provides a seamless range of gear ratios, improving efficiency by providing smooth gear transmission.

End Goal:

- To build a functional CVT model.
- Ensure the model is cost-effective and made using available workshop resources.

2. Plan of Action

Stage 1: Research and Design

- Study and finalize the transmission mechanism to be demonstrated.
- Develop detailed CAD models of all required components using Creo Parametric.

Stage 2: Material Procurement

- List required materials, ensuring total cost remains within PKR 5000.
- Procure materials based on manufacturing feasibility and strength requirements.

Stage 3: Fabrication Process

- Manufacture shafts, and base components using workshop tools.
- Ensure proper tolerances and alignments for smooth movement.

Stage 4: Assembly

- Assemble the components, ensuring proper belt tension and pulley alignment.
- Integrate bearings and shafts for optimal functionality.
- Secure all parts firmly while allowing necessary flexibility.

Stage 5: Testing & Adjustment

- Check movement consistency and transmission efficiency.
- Adjust tension and alignments where necessary.
- Ensure smooth rotational motion and demonstrate speed variation

Stage 6: Documentation & Report Submission

- Compile project findings, challenges, and optimizations.
 - Present the final report including all details.
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3. Workload Distribution

TASK	Number Of Members Working on it
Research and Design	1 Member
CAD Modeling	1 Member
Material Planning and Procurement	2 Members
Assembly and Testing	All Members
Fabrication	All Members
Report Compilation	1 Member

4. Step-by-Step Project Execution

1. Conduct background research on CVT and finalize the design.
2. Develop detailed CAD models considering fabrication constraints.
3. Procure materials while ensuring budget compliance.
4. Begin fabrication:
 - Manufacture shafts.
 - Cut and refine support structures.
 - Drill necessary holes for mounting and alignment.
 - Assemble all parts with precision.
5. Conduct testing.
6. Finalize adjustments and document results.

5. Tentative List of Materials & Fabrication Methods

Item	Quantity	Material	Fabrication Method	Cost (PKR)
Shaft	1	Plastic / Wood	Turning	100
Rubber Belt	1	Rubber	Purchase	100
Bearings	2	Steel	Purchased	600
Wooden Base	1	MDF Wood	Cutting	-
Screws & Nuts	10	Steel	Purchased	50
Miscellaneous	-	-	-	1500
Total Cost	-	-	-	2350
